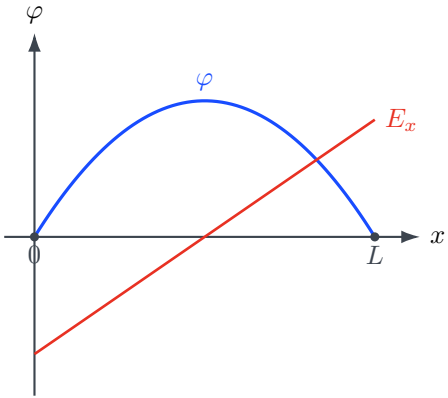
一维泊松方程：两端电势固定后，均匀正电荷产生下凹电势



$$\varphi(x) = \frac{\rho_0}{2\varepsilon_0} x(L - x)$$

$$E_x(x) = \frac{\rho_0}{\varepsilon_0}\left(x - \frac{L}{2}\right)$$

两解之差满足 $w'' = 0$ 且
$w(0) = w(L) = 0$，故 $w \equiv 0$。